St. Lawrence College

Tribhuvan University

Institute of Science and Technology

Final Year Project Report on

"ShopEase: Online Grocery Store"

Submitted to

Department of Computer Science and Information Technology

St. Lawrence College

In partial fulfillment of the requirements for the Bachelor's Degree in Computer Science and Information Technology

Submitted by

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SUPERVISOR’S RECOMMENDATION

I hereby recommend that this project work prepared under my supervision by below listed team of students entitled **“ShopEase: Online Grocery Store”** to be accepted as in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Information Technology be processed for the evaluation. In my best knowledge this is an original work in Computer Science and Information Technology.

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STUDENT'S DECLARATION

We hereby declare that we are the only authors of this work and that no sources other than that listed here have been used in this work.

…………………………………………

Ram Kumar Timalsina,

Sanjay Tripathi,

Sushant Pant

November 2024,

St. LAWRENCE COLLEGE

Tribhuvan University

LETTER OF APPROVAL

This is to certify that this project prepared by **Ram Kumar Timalsina, Sanjay Tripathi** and **Sushant Pant** entitled **“ShopEase: Online Grocery Store”** in partial fulfillment of the requirements for the degree of B.Sc. in Computer Science and Information Technology has been well studied. In our opinion it is satisfactory in the scope and quality as a project for the required degree.

……………………………… Mr. Parash Mishra Project Supervisor St Lawrence College	……………………………… Mr. Basanta Gyawali Coordinator St Lawerence College

ACKNOWLEDGEMENT

The success and outcome of this project required a lot of guidance and assistance from many people. We feel privileged to have received this during the development phase of our project. All that we have done is only due to such supervision and assistance. We would not forget to thank them.

We respect and thank Mr. Parash Mishra for providing us an opportunity to do the project and giving us all support and guidance, which made me complete the project duly. We are thankful to him for his gratifying support and guidance. We owe our deep gratitude to him for his keen interest in our project work by providing all the necessary information for developing this system.

We are thankful and fortunate enough to get constant encouragement, support, and guidance from all our friends and family members, which helped us complete the project work.

Ram Kumar Timalsina

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ABSTRACT

ShopEase makes online shopping easy and convenient. Unlike traditional shopping, which requires physical effort, e-commerce saves time and effort. ShopEase is a simple e-commerce website where customers can easily buy products.

ShopEase offers many benefits, such as browsing products, categorizing items, and ordering from home. This is especially useful for students and busy individuals. Our project aims to build a reliable e-commerce website using the MERN stack (MongoDB, Express.js, React, and Node.js). This technology stack helps create a dynamic and responsive site that can handle high traffic and provide a smooth user experience. Initially, we plan to sell various grocery products. The project focuses on making grocery shopping straightforward and efficient, especially for people with busy lives. It also aims to make grocery prices more transparent for everyone.

Keywords: *Ecommerce; Electronic Payment Methods; User Experience; Direct Online*

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LIST OF ABBREVIATIONS

API	Application Programming Interface
B2C	Business-to-Consumer
LAN	Local area network
PWA	Progressive Web App
SDLC	Software Development Life Cycle
VSCode	Visual Studio Code

CHAPTER 1: INTRODUCTION

1.1 Overview

Electronic commerce or ecommerce is a term for any type of business, or commercial transaction that involves the transfer of information across the Internet. It covers a range of different types of businesses, from consumer-based retail sites, through auction or music sites, to business exchanges trading goods and services between corporations. It is currently one of the most important aspects of the Internet to emerge.

The consumer moves through the internet to the merchant's web site. From there, he decides that he wants to purchase something, so he is moved to the online transaction server, where all of the information he gives is encrypted. Once he has placed his order, the information moves through a private gateway to a Processing Network, where the issuing and acquiring banks complete or deny the transaction. This generally takes place in no more than 5- 7 seconds.

There are many different payment systems available to accommodate the varied processing needs of merchants, from those who have a few orders a day to those who process thousands of transactions daily. With the addition of Secure Layer Technology, E-Commerce is also a very safe way to complete transactions

Thus, this paper describes an e-commerce system that will make user easy to buy groceries and easy payment method to pay.

1.2 Background and Motivation

Nepal is a developing country where Information Communication and Technology (ICT) plays an essential role in its growth. E-commerce, defined as buying and selling products or services over electronic systems like the Internet, is steadily gaining traction in Nepal. Although online payment options are still limited, new services such as Esewa and Khalti are beginning to revolutionize Nepal's e-commerce landscape, making it easier for businesses to trade online and for consumers to access products. With these platforms, grocery stores and other retailers have new opportunities to reach customers and expand their businesses through digital means.

E-commerce is quickly becoming an accepted business model in Nepal, with more companies setting up websites to offer products and services online. Shopping from the convenience of one's home is becoming common, even for essential items like groceries.

The objective of this project is to develop a user-friendly online grocery store where customers can purchase a wide variety of items, from fresh produce to packaged goods and household essentials, all from the comfort of their homes. This store will be equipped with a virtual catalog, allowing customers to browse various grocery items, add them to their shopping cart, and place orders seamlessly.

In this virtual grocery store, customers can explore categories such as fruits and vegetables, dairy products, pantry essentials, and beverages. Once they select items, they can review their shopping cart at checkout. To complete the purchase, customers will provide details like billing and shipping addresses, delivery preferences, and payment options. Convenient online payment methods, such as digital wallets, bank transfers, and cash on delivery, will be available to cater to customer needs.

This online grocery store will simplify shopping for everyday essentials, saving customers time and effort while promoting the growth of e-commerce in Nepal.

1.3 Problem Statement

E-commerce provides a convenient platform for selling products to a large customer base. However, competition is high among numerous online grocery stores, and customer expectations are equally high. When users visit an online grocery site, they expect to find what they need quickly and efficiently. Often, customers may not know the exact brand or type of grocery product they want; they may have only a general idea, like “fresh vegetables” or “snacks for the week.”

Many customers also start their shopping journey on search engines like Google, expecting to be directed to online stores that offer the groceries or products they are seeking. This behavior highlights the importance of helping customers easily navigate a grocery store’s offerings to narrow down their broad ideas and finalize their purchases.

The purpose of an online grocery website is to guide customers from a general search to specific product selections that meet their needs. For instance, if a customer is interested in buying fresh vegetables, the website should provide options to filter by categories. As the customer applies filters based on their preferences, they will be presented with a refined list of options, making it easier to choose the exact products they want.

By offering these filtering options, an online grocery store can simplify the decision-making process, allowing customers to find the groceries they need quickly. This streamlined experience encourages customers to complete their purchases, helping the store build customer satisfaction and loyalty.

1.4 Objectives

The objectives of this project are:

- To make distraction free online grocery site.
- To get and maintain loyal users as making stable relationship with your existing customers is significant.
- To Enhance the Efficiency of Services by opting for the online E-commerce platform.
- To implement a recommendation algorithm for an enhanced user experience.
- To discourage use of cash with the integration of online payment method.

1.5 Scope and Limitation

The scope of online grocery shopping is growing rapidly due to the increasing number of internet users worldwide. Consumers are spending more time shopping online for a range of daily essentials available on e-commerce platforms. Online grocery stores, in particular, benefit from seasonal promotions and discounts, which attract customers looking for convenient ways to save on household necessities. This trend is especially promising in developing countries.

In Nepal, the expanding internet user base makes e-commerce, including online grocery stores, a sustainable and profitable business model. The potential for a long-term, trending marketplace for groceries in Nepal is substantial, providing opportunities for online stores to reach a wider customer base.

1.6 Key Areas of Focus

- **Customer Service**

  Excellent customer service is essential for success in the online grocery business. Providing prompt delivery and seeking feedback after each order can enhance customer satisfaction. Whether selling groceries on a marketplace platform or an independent website, offering high-quality customer service is critical. Regular feedback collection can help understand customer needs and satisfaction, creating a personalized shopping experience.

- **Expanding Reach**

  One of the biggest advantages of an online grocery store is its ability to reach customers beyond local boundaries. Customers can visit the website from anywhere, order groceries, and have them delivered to their doorstep. With people around the world increasingly adopting online grocery shopping, an online store can cater to a diverse and widespread audience.

1.7 Limitations of an Online Grocery Store

- **Tax Issues**

  Sales tax can be a complex issue for online stores, especially when dealing with orders from different regions. This can impact the pricing and may create challenges for online grocery stores to maintain competitive rates with local stores, particularly if tax exemptions vary.

- **Cultural Obstacles**

  Online grocery stores serve customers from diverse backgrounds, each with unique shopping habits, preferences, and dietary restrictions. Adapting to these differences and addressing language barriers can sometimes create challenges between the seller and buyer.

- **Legal Compliance**

  An online grocery business must comply with a range of legal requirements, which may vary by country or region. These regulations, including food safety and e-commerce laws, are essential for the business but can present hurdles for expanding online grocery operations.

- **Privacy Concerns**

  Customer privacy remains a concern in online shopping. Online grocery stores must collect personal details like addresses and phone numbers for delivery, which can raise privacy concerns among customers. Ensuring secure handling of customer information is critical to building trust, but not all online platforms have the advanced technology needed to fully protect sensitive data.

1.8 Development Methodology

The **Waterfall Model** was the first software development process model introduced and is also known as a **linear-sequential life cycle model**. This model is straightforward and easy to understand. In the Waterfall Model, each phase must be fully completed before moving on to the next, ensuring a sequential, step-by-step approach with no overlap between phases.

As the earliest **Software Development Life Cycle (SDLC) model** widely used in software engineering, the Waterfall approach helped standardize project success. The model illustrates the software development process as a linear sequence, where each phase's output serves as the input for the subsequent phase, progressing from start to finish. [1]

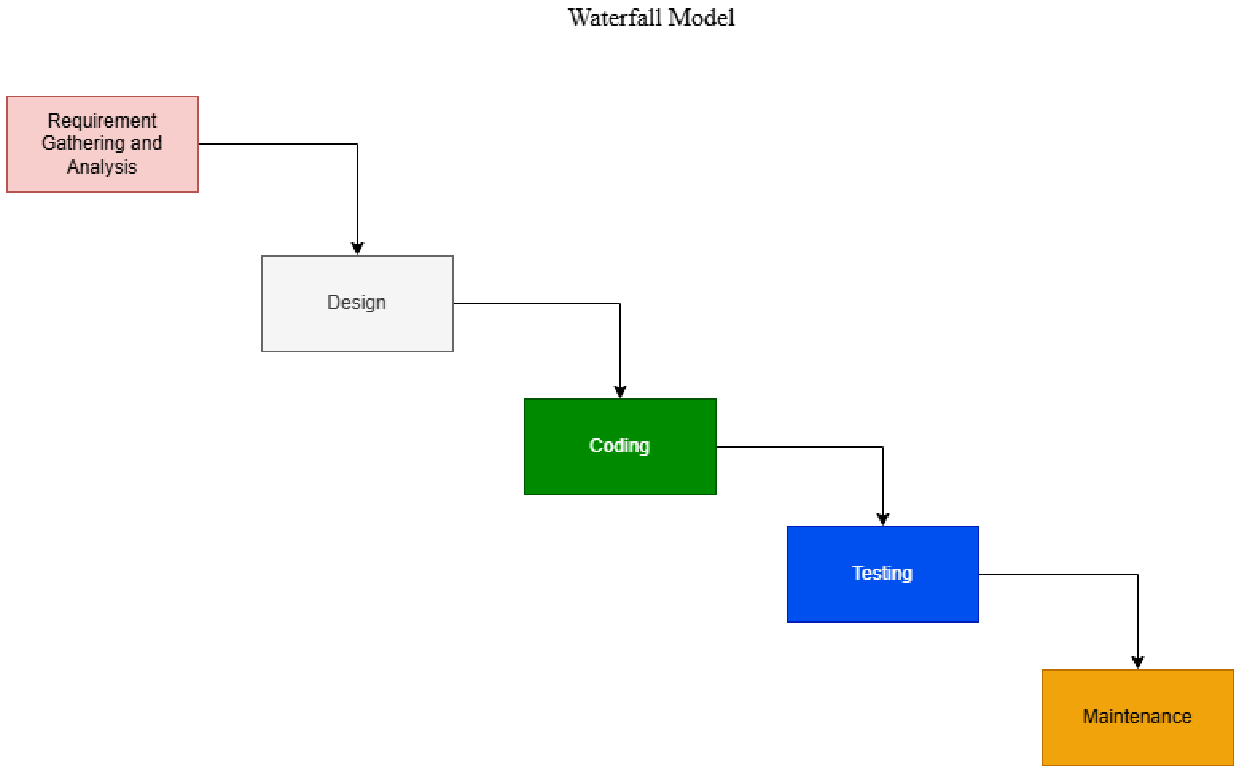

Figure 1.6: Waterfall model

1.9 Outline of the Report

Chapter 1: Introduction gives a general overview of the project along with the background and motivation, problem statement, objective, scope, limitation and the methodology used. Overview provides a basic information about the inner workings of the program. Background and motivation describe how visualization aids better understanding and incentives that led to the development of the program. The problem statement addresses the question "What is the problem that the project will address?" Objectives claims the statements that outlines the problem addressed by the project. The scope defines the area and the information included in the project. The limitation defines the areas in which the project should work on. The methodology defines which software model is used to conduct the project.

Chapter 2: Background Study and Literature Review includes background study, literature review, the current system being used and the problems concerned with it. Background study describes the fundamental theories and concepts related to the project. Literature review highlights the importance of visualization in learning algorithms.

Chapter 3: System Analysis includes requirement analysis, functional requirements, non-functional requirements, feasibility analysis and the analysis. Requirement analysis provides the descriptions of what the system should do; the services it provides and the constraints on its operation. Functional requirements describe the functionality of the system whereas the non-functional requirements describe the system properties related to system performance and usability. Feasibility analysis is an analysis that considers all the project related factors. The analysis may involve the UML diagram based on structured or object-oriented approach.

Chapter 4: System Design includes a design and algorithm. Design is the design of UML diagram which is involved in the project. Algorithm is the process which is used to develop the project.

Chapter 5: Implementation and Testing presents the tools and technology used for the development of the project, implementation and testing. Implementation describes the system at a finer level of detail, down to the code level. Testing provides the techniques that is used to remove the bugs from the project.

Chapter 6: Conclusion and Future Recommendation Conclusion gives a summary of the entire project and proposes question for further study and aims for broader implications. Future Recommendation gives the future work that can be added to the project to make it more feasible

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background study

E-commerce was one of the first industries to harness the power of recommendation system and data analysis. Today, there are countless applications that improve various aspects of e-commerce, from managing inventory to providing customer service. One crucial element in e-commerce is the recommendation engine, which helps boost profits and attract new customers.

Moreover, Recommendation system depends on set of data. Depending on the data set distinction is created between different types of system. Typically, content-based and collaborative system are used in E-commerce. The system we are going to develop using the MERN stack (MongoDB, Express.js, React, and Node.js) is featured by content-based recommendation system. Many websites use content-based recommendation system.

The (Grocery Store) built on MERN stack offers a full-stack JavaScript solution, enabling developers to work seamlessly across both the front-end and back-end of the application. This technology stack is well-suited for developing dynamic and responsive e-commerce sites that can handle high traffic and provide a smooth user experience.

2.1.1 Recommendation engine (recommender system)

The recommendation engine is a critical tool in e-commerce, focusing on personalization and tailored user experiences. By analyzing vast amounts of data related to user interactions and preferences, businesses can create product recommendations that align closely with individual customer interests. [2]

2.1.2 How does the recommendation engine work in e-commerce?

The recommendation engine in e-commerce functions by analyzing the collected data on user interactions with the website. This analysis can reveal which products a customer viewed, what they searched for, and where they spent the most time. By utilizing various factors—such as a customer's previous activity, their favorite products, and contextual information like location and weather—the system can generate personalized product suggestions that are likely to interest them.

For instance, if a customer frequently searches for specific items, the recommendation engine stores these interests in the user's local storage. This information is then used to suggest products that align with their previous searches, creating a tailored shopping experience. Furthermore, the system can also incorporate the purchasing behaviors of other customers in the same region, enhancing the relevance of the recommendations. [3]

2.2 Literature Review

The amount of information in e-commerce is increasing far more quickly than our ability to process. Recommender systems apply knowledge discovery techniques to help people find what they really want. However, all of the previous approaches have an important drawback: items added newly cannot be found. In this paper, a general framework is proposed for supporting automatic recommendation of the new item to the potential users based on the concept of influent sets. We propose a simple efficient indexing structure and a heuristic information retrieval technique algorithm for searching reverse k nearest neighbor in high-dimensional dataset. And experimental evaluation reveals that our approach outperforms the previous algorithm and enhances the performance efficiently.

2.3 Study of Existing System

While researching about such sorts of project. I found out some of the project which is available for web are described below:

Amazon

Amazon ML supports three types of ML models: binary classification, multiclass classification, and regression. The type of model you should choose depends on the type of target that you want to predict. ML models for binary classification problems predict a binary outcome (one of two possible classes). To train binary classification models, Amazon ML uses the industry-standard learning algorithm known as logistic regression. ML models for multiclass classification problems allow you to generate predictions for multiple classes (predict one of more than two outcomes). For training multiclass models,

Amazon ML uses the industry-standard learning algorithm known as multinomial logistic regression. ML models for regression problems predict a numeric value. For training regression models, Amazon ML uses the industry-standard learning algorithm known as linear regression. [3]

Daraz

It is the fastest growing E-Commerce platform in Nepal offering an unparalleled shopping experience at the comfort of your home. You can enjoy convenient and hassle-free shopping without stepping out of the house. Daraz Nepal is a company fully owned by Alibaba Group and runs e-commerce platform to provide wide-range of products along with groceries under a single roof. To enhance the online shopping experience, Daraz provide customers with tools that help make the best purchase decision. Daraz provide various platforms to bridge the gap between sellers and buyers. Any query about the product can be asked directly to the seller through instant messaging. [4]

Kirana

Kirana provides services to those who want to purchase groceries through online medium. It is the place where user can buy groceries at reasonable prices. It also promises to provide same day delivery. User can also create shopping list. It has got some good feature such as Categorization, provide delivery within same day etc. Similarly it possess some lacking features such as option to rate is absent ,no proper inventory detailing(stock),etc.

SalesBerry

SalesBerry, established in 1993, has its head office in Sadobato, Lalitpur. Currently, SalesBerry is a well-known departmental chain store in Nepal that also offers online services. The design is impressive and provides a wide range of options to choose from. [5]

1.4 The problem with current system

The current system consists of few apps that provide basic features such as.

1. Product directory system with featured products.
2. Allows paid membership option for better functionalities.
3. These product management systems don’t provide better recommendation.

CHAPTER 3: SPECIFICATION AND DESIGN

3.1 Requirement Elicitation and Analysis

The application must be able to perform basic fundamental tasks such as listing prodcut in just one click. The system should provide authentic counts of add to cart and watchlisting of each products to the admin for better analytics. Users should be able to buy immediately or later depending on what they choose

3.1.1 Functional Requirement

A functional requirement specifies what the system should provide. For our system, the functional requirements are as follows

- The User shall be able to register and login
- The Application shall take Minimum steps to make a purchase
- The Application shall be Web-compatible
- The Application shall be Accessible

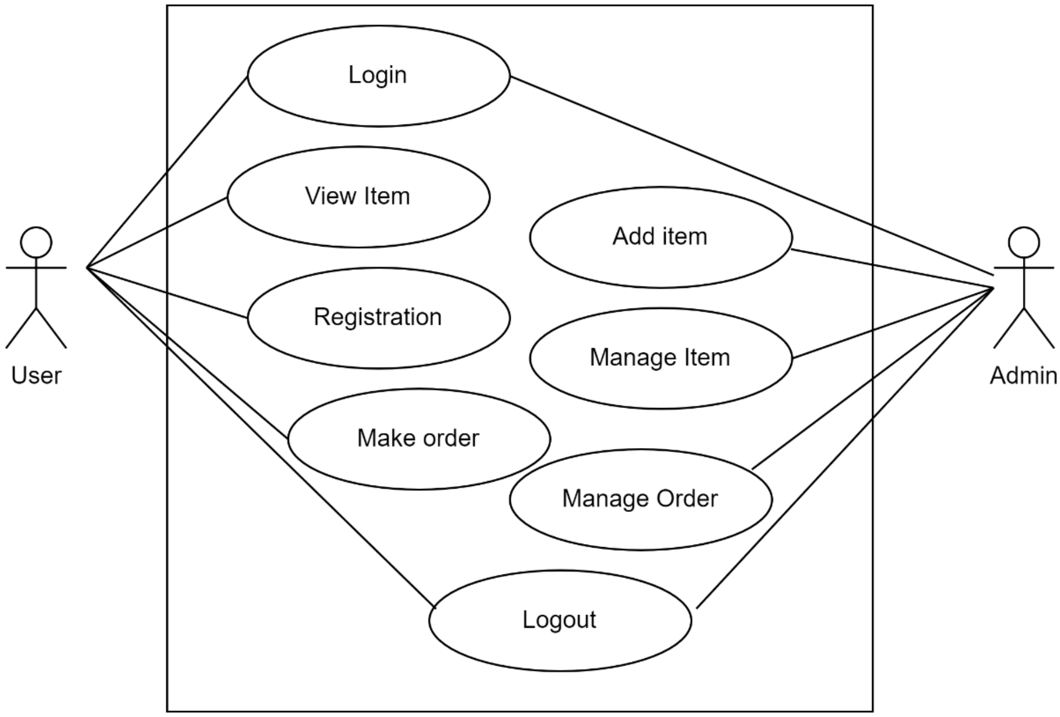

Figure 3.1: Use Case Diagram of Ecommerce Application

3.1.2 Non-Functional Requirement

Non-functional requirements define the overall qualities or attributes of a system. So, the non-functional requirements for our system are as follows:

- The system must be a user-friendly UI / UX.
- The system must be responsive.in design.
- Response time must be minimal.
- Web and database security must be high.
- Sever error as well as client-side error must be logged and understandable.
- REST endpoints must be precise and efficient to document.

3.2 Feasibility Analysis

3.2.1 Technical Feasibility

In this project, the system was implemented as a web-based application and it runs all systems. The system uses JavaScript's web framework – Node and Express for the backend. This system uses JavaScript's web framework – React JS for the frontend. The recommendation system is used using a content based recommendation algorithm

3.2.2 Operational Feasibility

The system built in the project follows a client-server architecture. This project will have responsive interface and the interaction between user and system will be through PC"s, tablets as well as mobile devices. Thus, this system can be operated by guest as well as logged in users.

3.2.3 Economic Feasibility

Each and every requirement for the project is available online and through various research papers, journals, no external expenses are needed.

3.2.4 Schedule Feasibility

The project is estimated to be completed in 4 months with the first prototype of the project developed within the 1st month

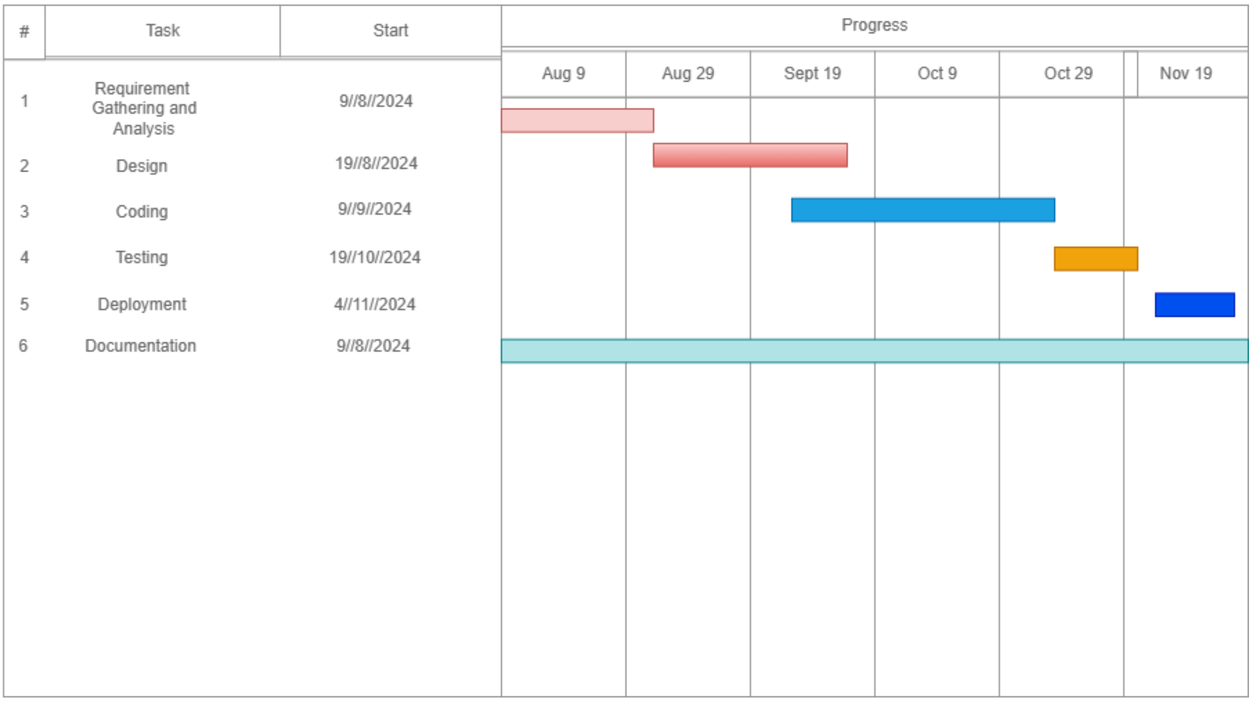

Figure 3.2 : Gantt Chart of project lifecycle

3.3 Analysis

The figure below are the sequence diagram and activity diagram of the project. When we start the application, we are directed to the home page. The home page contains all the products available and filter section along with navigation section. In the navigation section we have different pages path along with search bar.

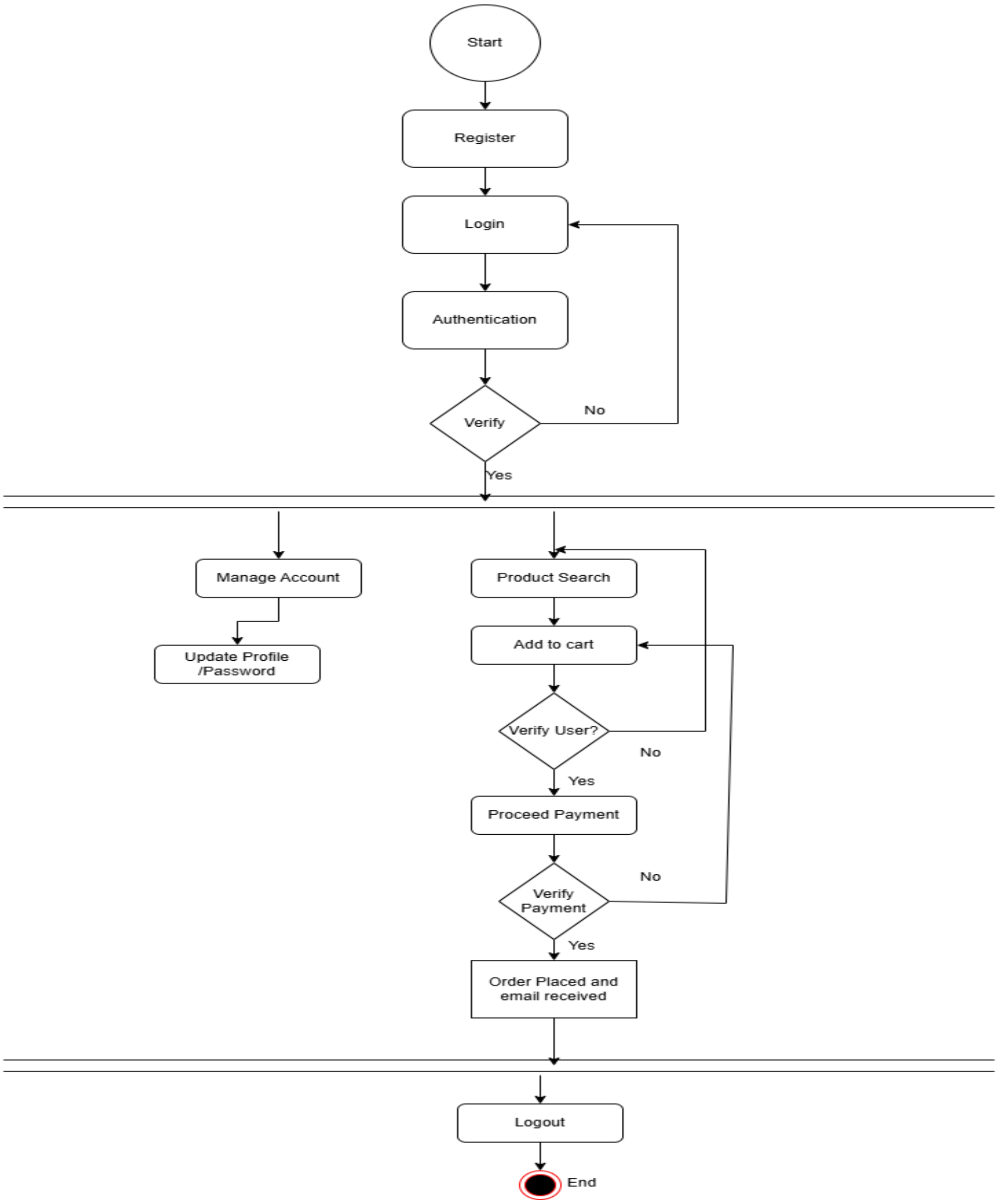

Figure 3.3.1 : User Activity Diagram of Ecommerce

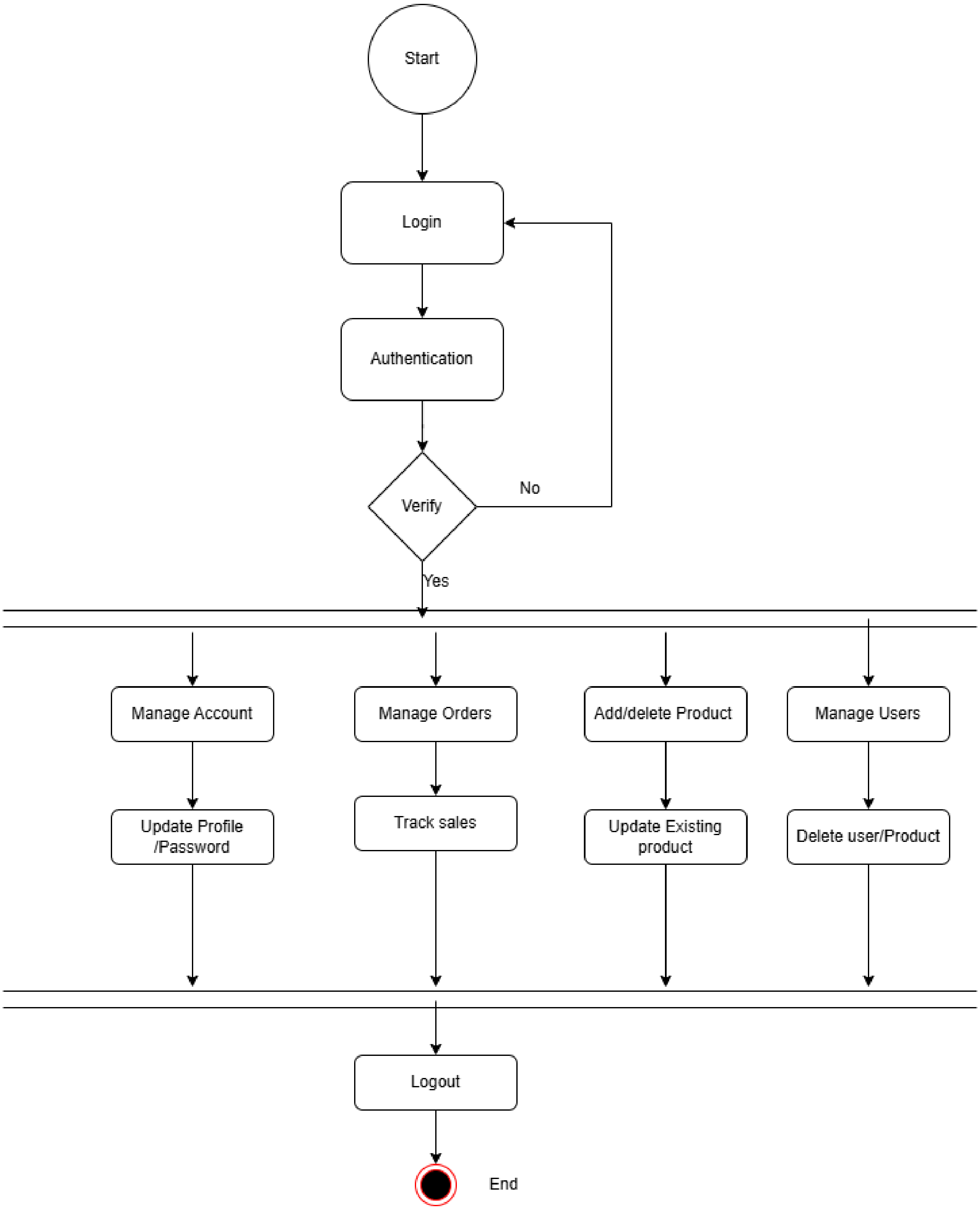

Figure 3.2.2 : Admin Activity Diagram of Ecommerce

Below is the detailed sequence diagram of the project:

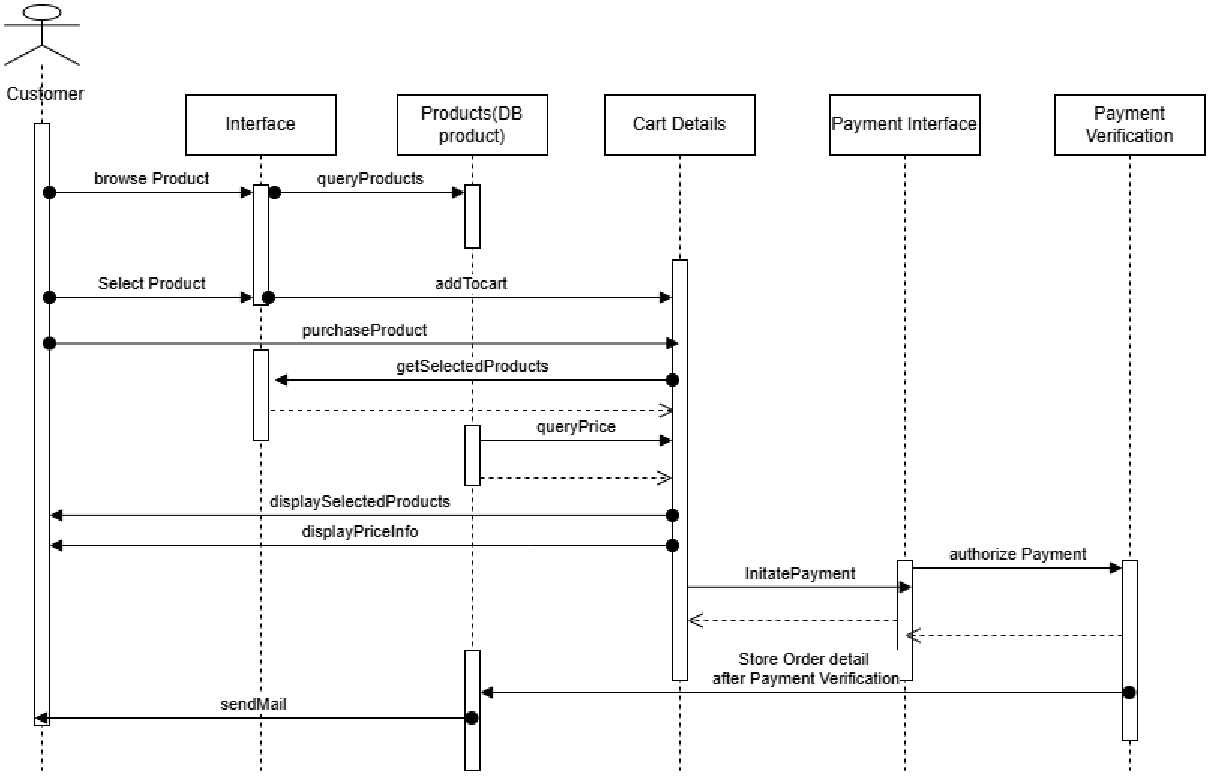

Figure 3 3.3: Sequence Diagram of Ecommerce

CHAPTER 4: SYSTEM DESIGN

4.1 Design

4.1.1 Deployment Diagram

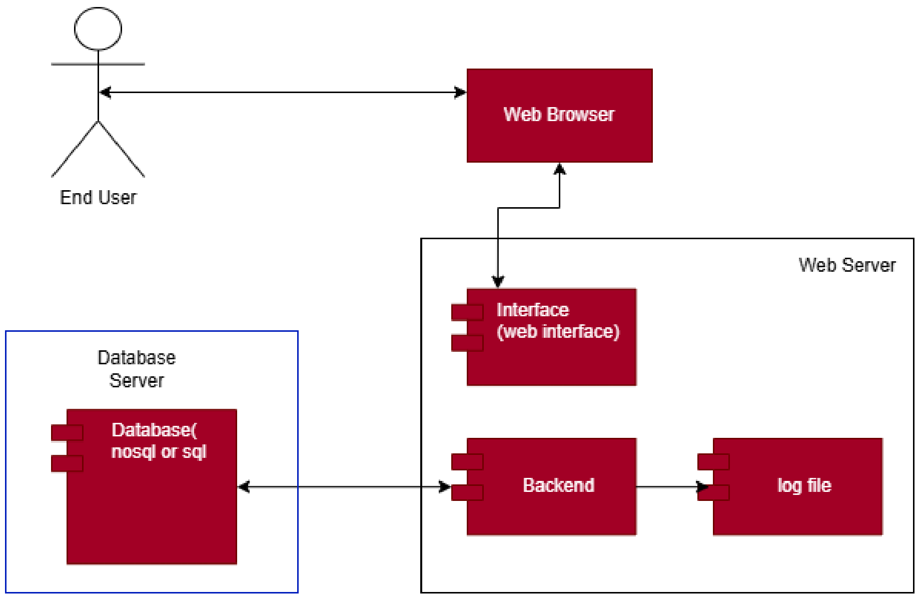

Figure 4.1.1 Deployment Diagram of Ecommerce Website

4.1.2 Component Diagram

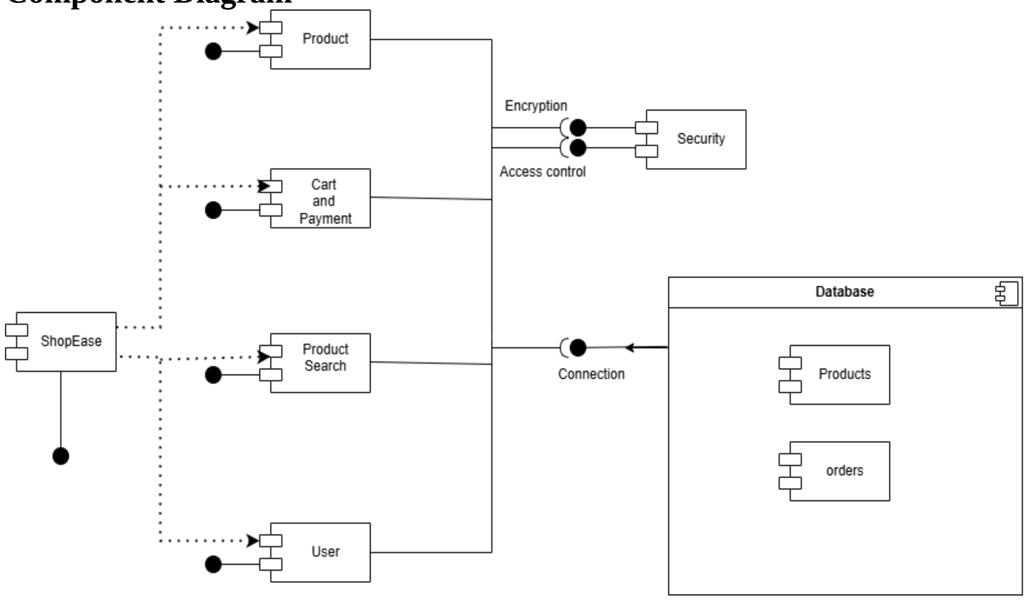

Figure 4.1.2 Component Diagram of Ecommerce Website

4.2 Algorithm Details

The following algorithms were used to build the system

4.2.1 Content Based Recommendation Algorithm

The recommendation system utilizes a content-based algorithm, which tailors suggestions to each user based on their search and browsing history. This algorithm collects and analyzes individual user searches, using these patterns to recommend related items within similar categories. For example, if a user searches for "milk," the algorithm will suggest other dairy products or similar items that belong to the same category as milk.

Content-based recommendation systems operate by identifying patterns in the user's interactions and leveraging item features to align suggestions with user preferences. Unlike collaborative filtering, which relies on similarities across users, this approach is purely user-centric, focusing solely on the individual's previous interactions. This makes it particularly effective for generating personalized recommendations without needing data from other users, enabling immediate and accurate suggestions based on recent activity. [6]

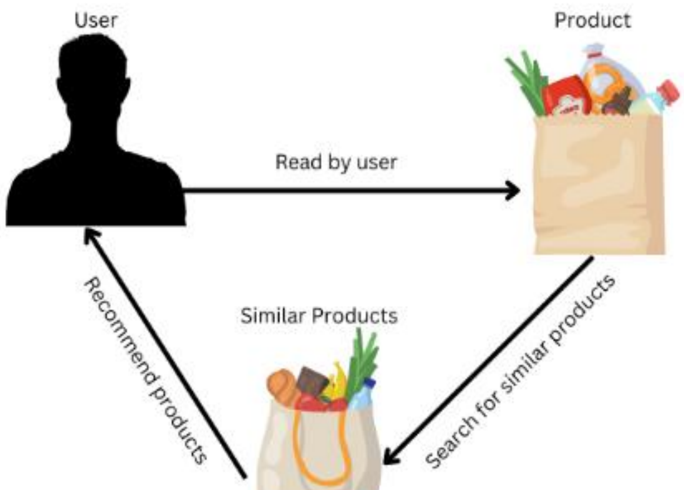

Figure 4.2.1 Content Based Recommendation

CHAPTER 5: IMPLEMENTATION AND TESTING

5.1. Implementation

The project was developed using one of the most emerging programming languages which has high value in the web development industry. JavaScript's Web Framework – React was used to develop the Server side and the client side of the project. The project uses REST API for the connectivity of the server and the client. Since these languages supports deployment on multiple platforms, the application was made to run on any browser of any platform.

The development of this project started with the development of wireframes for the application on every end. The initially the UI and Create, Read, Update, Delete (CRUD) operations for the application through the development of backend server was completed. Then we combined the two and implemented a recommendation algorithm.

5.1.1. Description of Important Files

searchFeature.js

The SearchFeatures class is designed to streamline search, filtering, and pagination for MongoDB queries in a Node.js application. It takes a MongoDB query object and a queryString of parameters, enabling flexible database querying based on user inputs. The search method checks if a keyword is provided in queryString and, if so, performs a case-insensitive search on the name field using MongoDB's $regex for partial matches. The filter method refines the query further by removing non-filter parameters (keyword, page, limit) from queryString, then converts comparison operators (like gt for greater than) into MongoDB's syntax (e.g., $gt). This allows for filtering based on fields such as price or ratings. Finally, the pagination method calculates the current page and skips the appropriate number of items to limit the results to the specified resultPerPage, effectively dividing the results into pages. This class is then exported as a module to be easily reused throughout the application, making the search process more efficient and organized.

5.1.2. Tools used

The list of all the tools used in the system development are mentioned below:

- **Back End: Express (Node.js Framework)**

Express is a minimal and flexible Node.js web application framework that provides a robust set of features for web and mobile applications.

- **Database: NoSQL (MongoDB)**

NoSQL databases emerged in the late 2000s as the cost of storage dramatically decreased. Gone were the days of needing to create a complex, difficult-to-manage data model in order to avoid data duplication.

MongoDB is a schema-less NoSQL document database. It means you can store JSON documents in it, and the structure of these documents can vary as it is not enforced like SQL databases. This is one of the advantages of using NoSQL as it speeds up application development and reduces the complexity of deployments.

- **UI Design: React**

React is web framework of Java Script for development of web application UI.

- **IDE Environment: VS Code, MongoDB Compass**

VS Code is used for developing React application and Express.

MongoDB Compass is a powerful GUI for querying, aggregating, and analyzing your MongoDB data in a visual environment. Compass is free to use and source available, and can be run on macOS, Windows, and Linux.

- **CASE tools: draw.io**

Draw.io is a web application which helps to create different diagram easily and effectively. The diagrams used in this report to illustrate the functionality of the project were made via draw.io.

5.2 Testing

5.2.1. Test Cases for Unit Testing

Table 5.2.1 : Test Case for Unit Testing

ID	Test Scenario	Test Case	Test Steps	Expected Result	Actual Result	Status
T001	Verify Suggested Section	To check wheatear the Suggestion section is displayed to the user or not	1. Open the application and see whether the selection section is there or not	User must be shown some amount of suggestion in the suggestion section if they have interacted with the application for certain time	The selection section is working fine for those user who is using this application	True
T002	Verify Suggested Items	To check if the item suggest to the user is correct of not	1. Open the application 2. Look at the suggested box	Always have the right suggestion according to your need	Products are displayed based on user interests	True
T003	Verify Button Functionality	To check wheatear the button is clickable or not	1. Open the Application 2. Click Button	User should be redirected to the main page	User gets redirected to the main page	True
T004	Verify recommendation generation	To check if recommendation is generated or not.	1) Open the app. 2) Select any products. 3) Look if products you may like is there	If recommendation n exists, it should be shown to the user	Recommendation was generated.	True

T005	Verify Create Products	To check if products are created or not.	1. Open the website as admin 2. Click on admin dashboard 3. Add Product 4. Add the necessary details	The product should be shown in the respective category.	Products are listed in the Product Page.	True
T006	Verify search box	To check if it allows to write or not	1) Open the app. 2) Click on search box	User should be allowed to type query.	User should be allowed to type query.	True

5.2.2 System Testing

Table 5.2.2 : System Testing for Ecommerce Application

ID	Test Scenario	Test Case	Test Steps	Expected Result	Actual Result	Status
T001	Verify Adding product	To check if new products can be added all the time	1) Open the app 2) Login to the app 3) go to admin dashboard 4)Add new products	New products should be added all the time when ever Admin adds the new product	Product can be added only by Admin	True
T002	Verify Editing Products	To check if the added products can be edited and change or not	1) Open the app 2) Login to the app 3) go to admin dashboard 4)Add new products	All the Section of the product should be able to be edited and those who aren’t edited should maintain their old value.	Product can be edited and unchanged product do retain their old value.	True
T003	Verify delete user	To check if the super admin and admin can delete the certain user or not	1) Open the app 2) Login to the app 3) go to admin dashboard	The admin should be able to be restrict or delete the User from the application	The admin is able to be delete the user from the application	True

			4)Add delete the user			
T00 4	Verify data loading	To check if data is loaded	1) Open the app 2) Login to the app. 2. Click Button	All the data should be loaded.	All the data is loaded when clicked	True
T00 5	Verify App Start	To check if the app starts	1) Open the app	The app should be opened	The app is opened	True
T00 6	Verify Products loading	To check if products are loaded	1.Open the app	To check if products are loaded	Products are loaded.	True
T00 7	Verify Recommendatio n	To check if recommendatio n is filtered.	1) Open the app 2) Select any podcast	Recommendatio n should be shown after certain time of using the app	Recommendatio n should be shown after	True
T00 8	Verify Categories loading	To check if categories are loaded.	1) Open the app	To check if Categories are loaded	Categories are loaded	True
T00 9	Verify Cart	To check if products are added in cart	1) Open the app 2) Add item 3) Check if it is in cart	To check if products are loaded in cart	Products are added in cart if clicked to add to cart	True

T00 10	Verify Checkout	To check user can checkout their selected product.	1) Open the app 2) go to cart page 3) click on check out if products are added in it	To check if Checkout works or not	Checkout works	True
T00 11	Verify Logout	To check if user can logout.	1) Open the app 2) If logged in click to log out	To check if user is logged out.	user is logged out.	True

5.3 Result Analysis

Nowadays, Ecommerce is really the name of the game**.** If you don't have an ecommerce website or online store to sell your products, you should create one right now. Not only is this a great way to sell to a larger audience, but you are also able to gain valuable data about which products interest them, which ad was the more successful, which search engine they used, and way more information about your audience's interests and behavior.

In the start of the application user aren't shown any suggestion as there is no record of the user search, item added to the cart etc. Once the user starts interacting with the application their data is recorded into the database. Data in a sense every type of data from the click of the product to the search of the product all are stored so that machine learning code can be applied to it Once there is enough activity the suggestion section starts popping up in the screen of the user for the suggestion giving algorithm we have used content based filtering Admin of the application can add product add new categories and they can even edit and delete the product and categories as per their requirement Overall functionality of the application runs

CHAPTER 6: CONCLUSION AND FUTURE COMMENDATION

6.1.Conclusion

E-commerce still represents one of the business methods that take advantage if done the right way, even if the stock market and commodities fell, but E-Commerce still able to survive and receive high transaction. E-Commerce has undeniably become an important part of our society. The successful companies of the future will be those that take E-Commerce seriously, dedicating sufficient resources to its development. E-Commerce is not an IT issue but a whole business undertaking. Companies that use it as a reason for completely re-designing their business processes are likely to reap the greatest benefits. Moreover, E-Commerce is a helpful technology that gives the consumer access to business and companies all over the world. The project has been constructed using MERN stack (MongoDB, Express(.js), React(.js) , Node(.js)) and content based recommendation algorithm for recommending and suggesting products to the user.

6.2.Future Recommendation

This application is equipped with latest technology and features which allows maximum scalability and updates in its near future. The application is built with REST API‟s which will allow the system to be shared with apps and other device software. Currently the content based recommendation to the user only. In the future we can add other Machine Learning model that may help the admin to boost their famous products that is more preferred by users and displayed them to non-authenticated users. Since it is a ecommerce site it will contain a hug number of images in future we can add image optimization techniques also.

In the future I have planned to:

- Collaborative filtering for recommendation system.
- Fine tuning the model to make it more accurate.
- Login through OTP authentication.

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APPENDIX

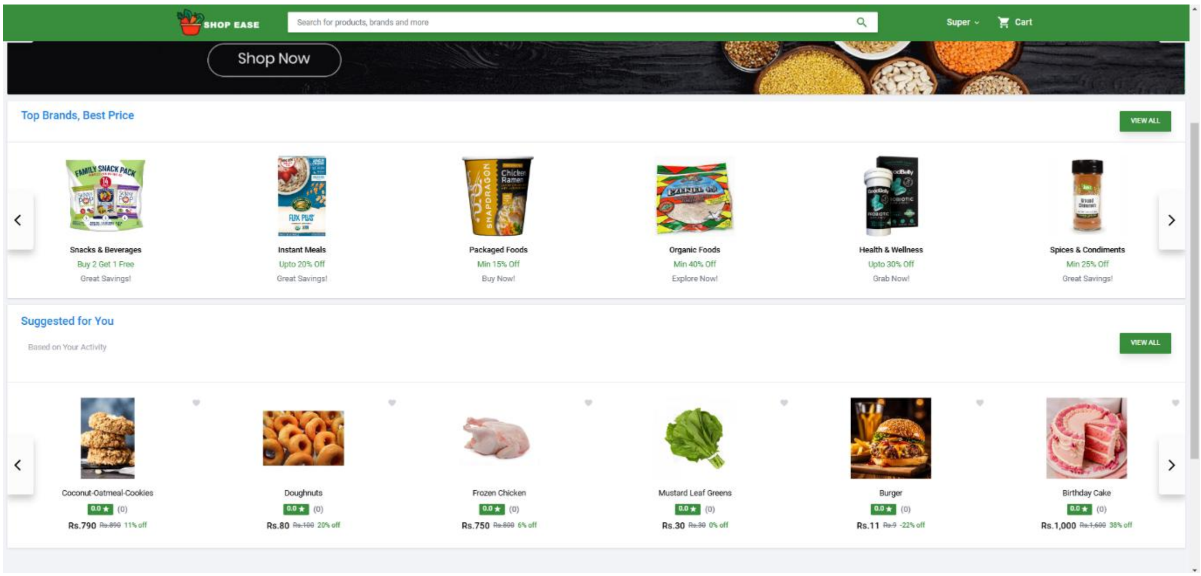


Figure A: Home Page Screen with Suggested Items

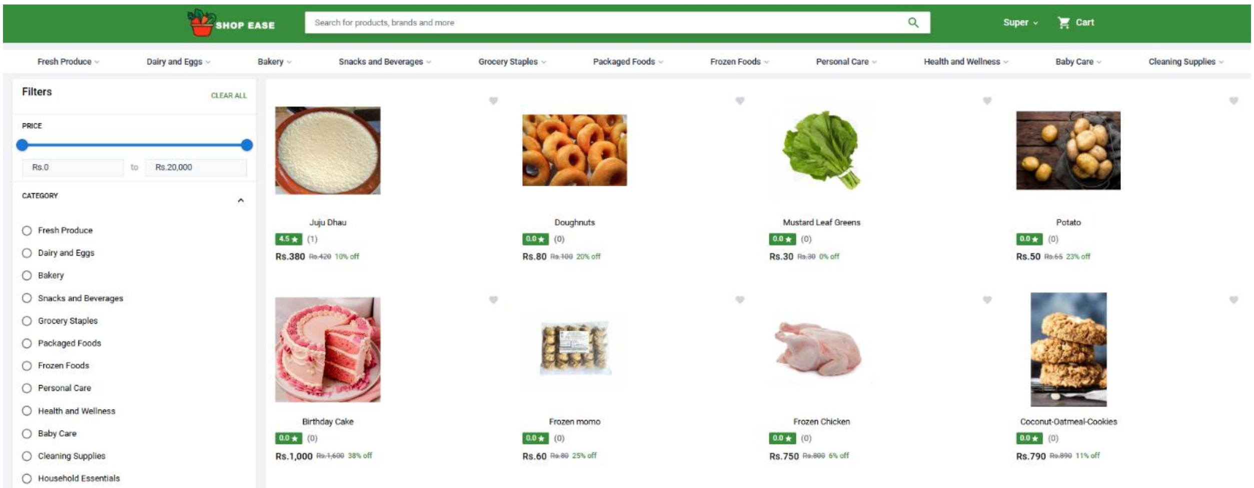
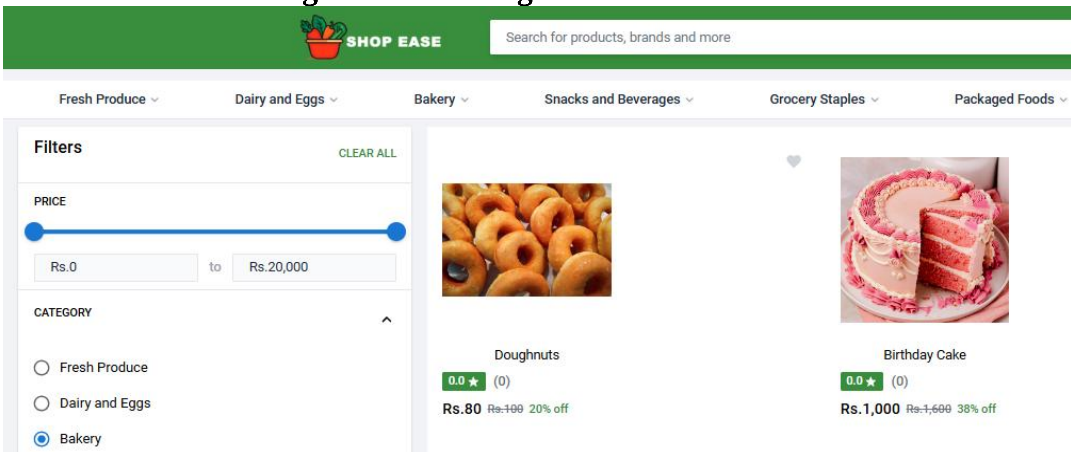
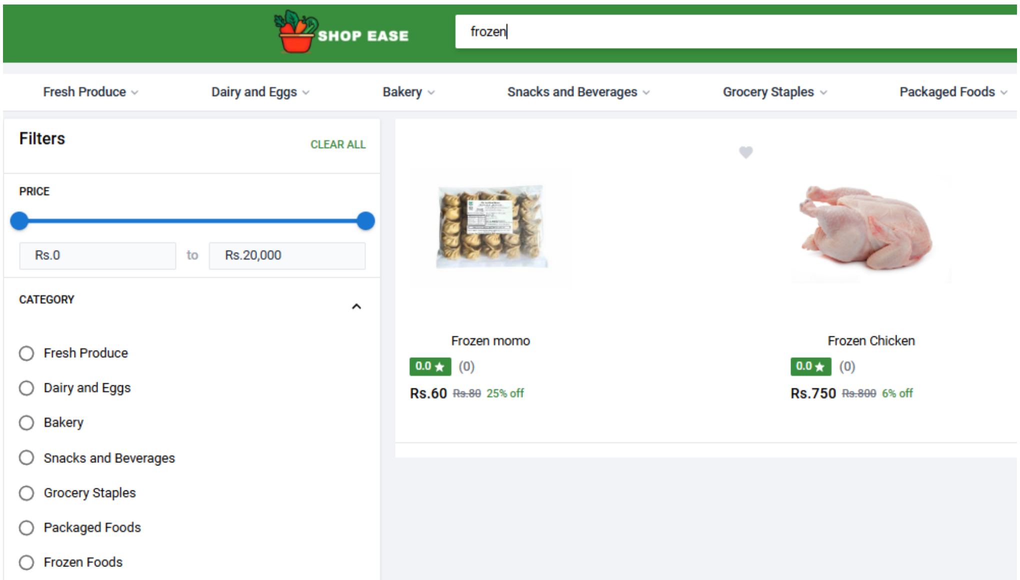


Figure B: Product Page Screen with all the Items

Figure C: Filtering Products



Filtering items with Category



Fitering items with Search

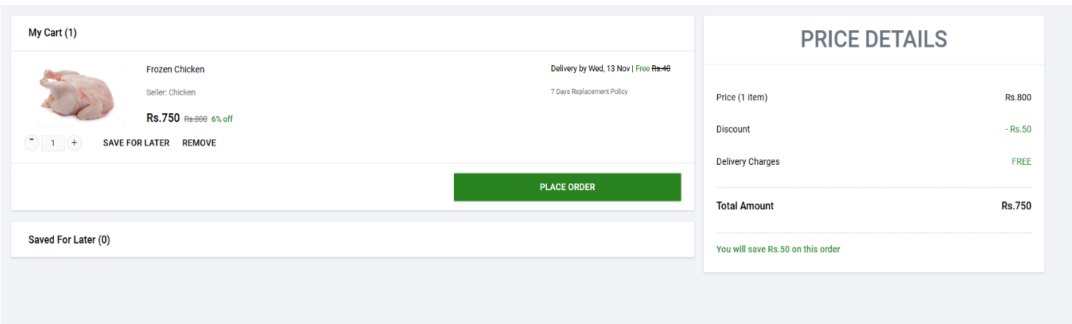


Figure D: Cart Screen with the Items Added to cart

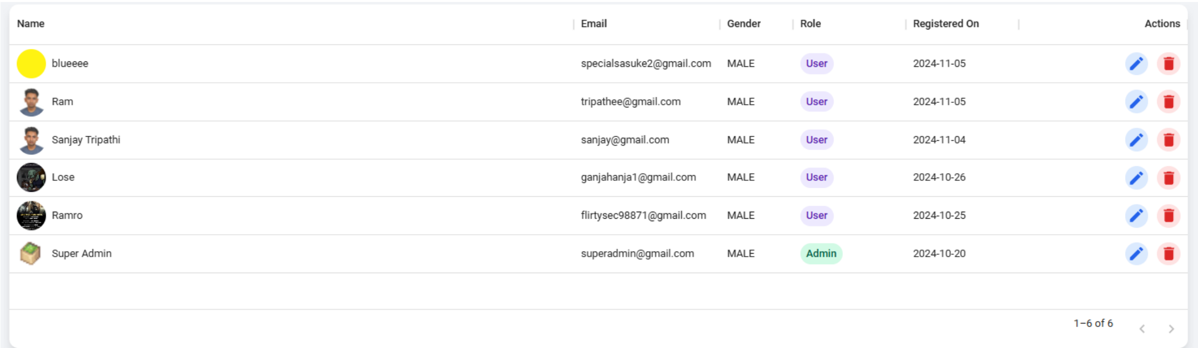


Figure E: Admin Screen with users displayed

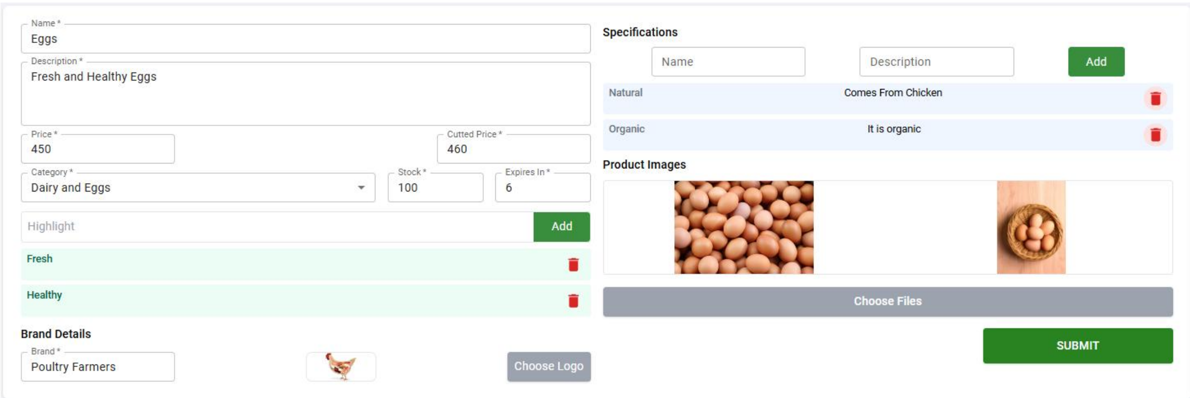
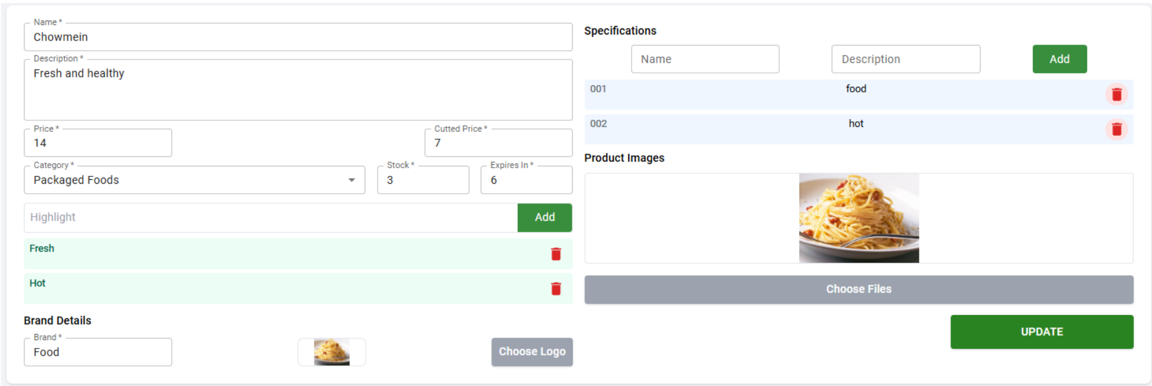


Figure F: Admin Screen To add product



Admin Screen To update product

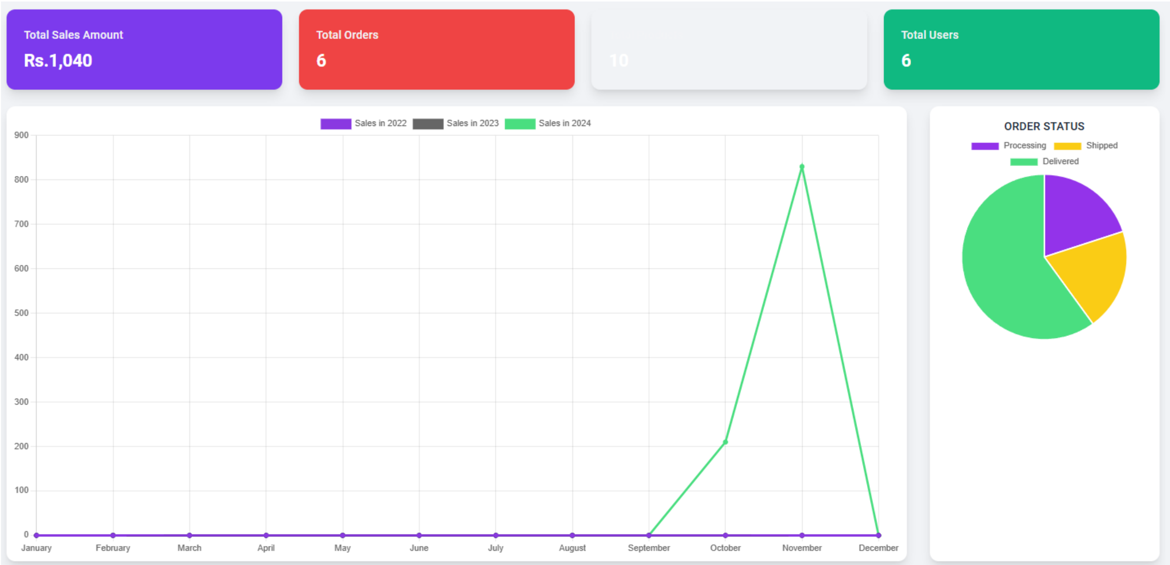


Figure G: Admin Screen to show chart

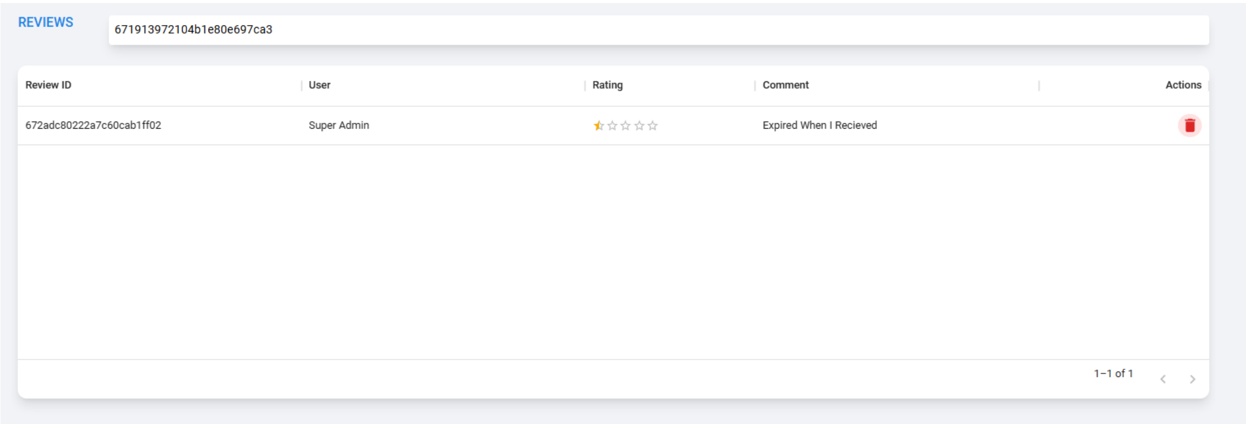


Figure H: Admin Screen to check reviews

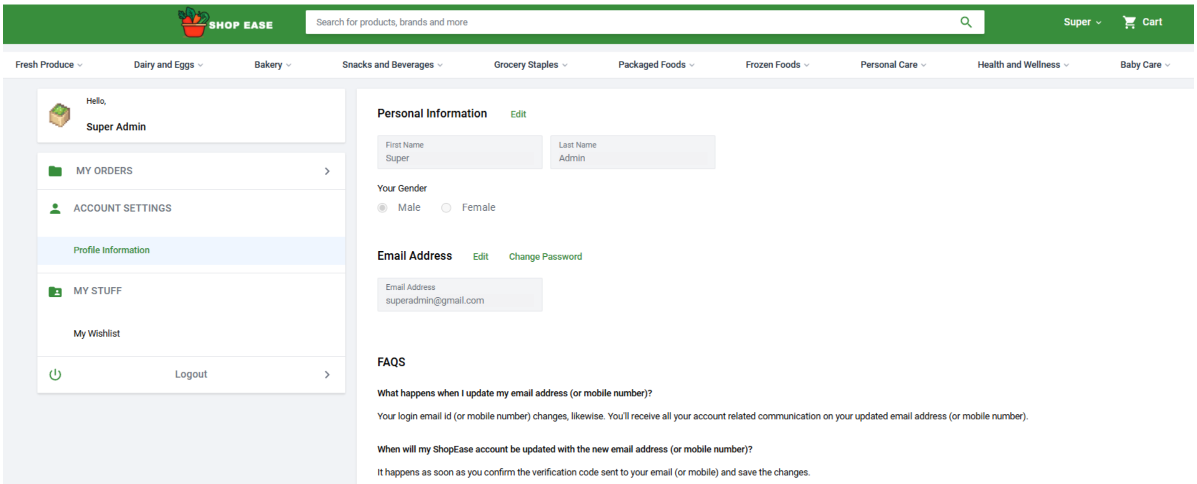


Figure I: User details displayed Screen

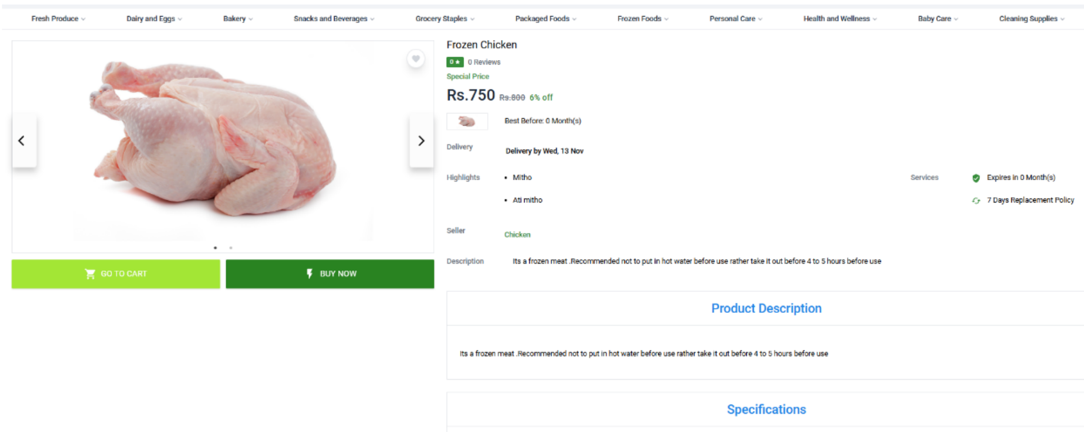


Figure II: Product Buy Screen

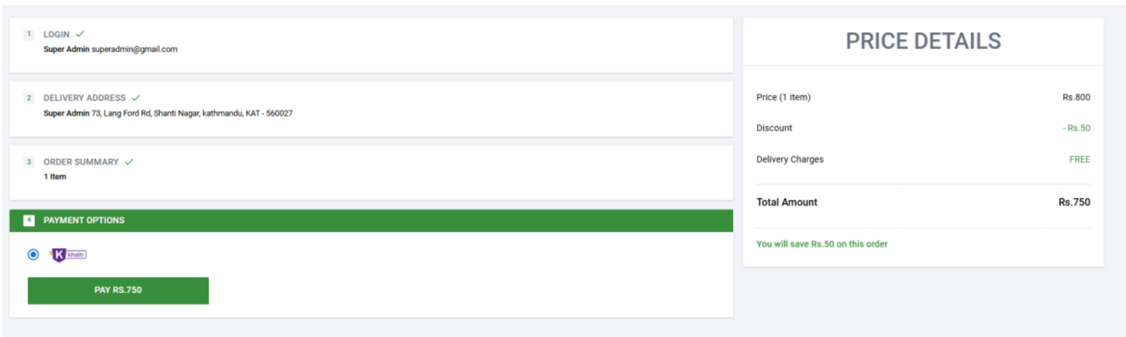


Figure III: Product Order Screen

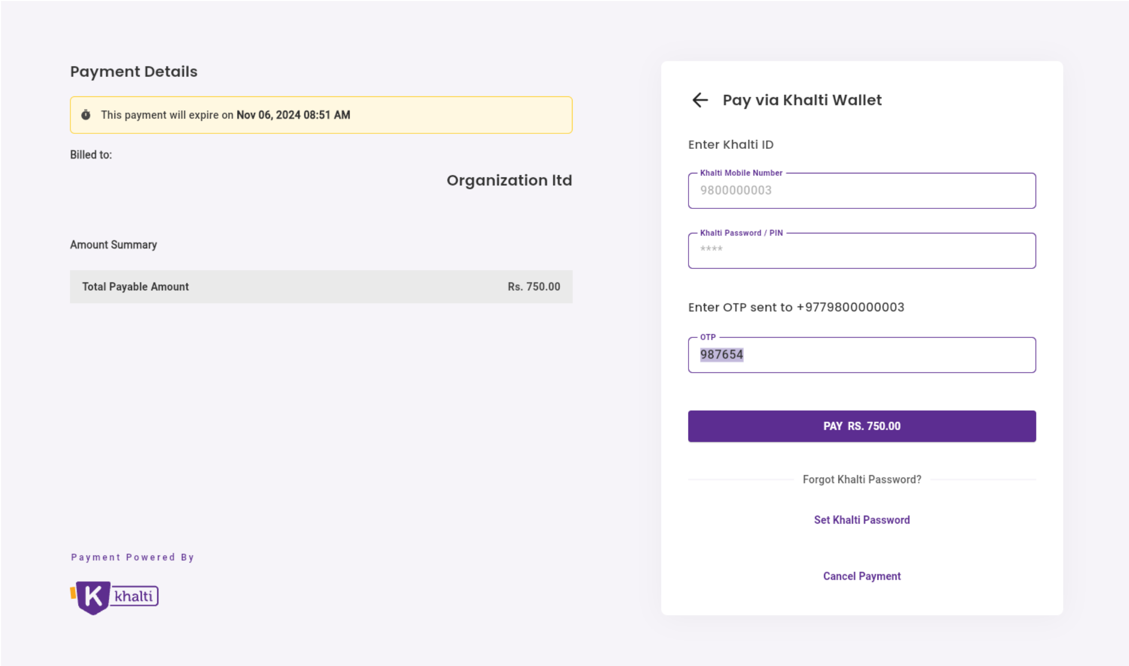


Figure IV: Khalti Payment Integration

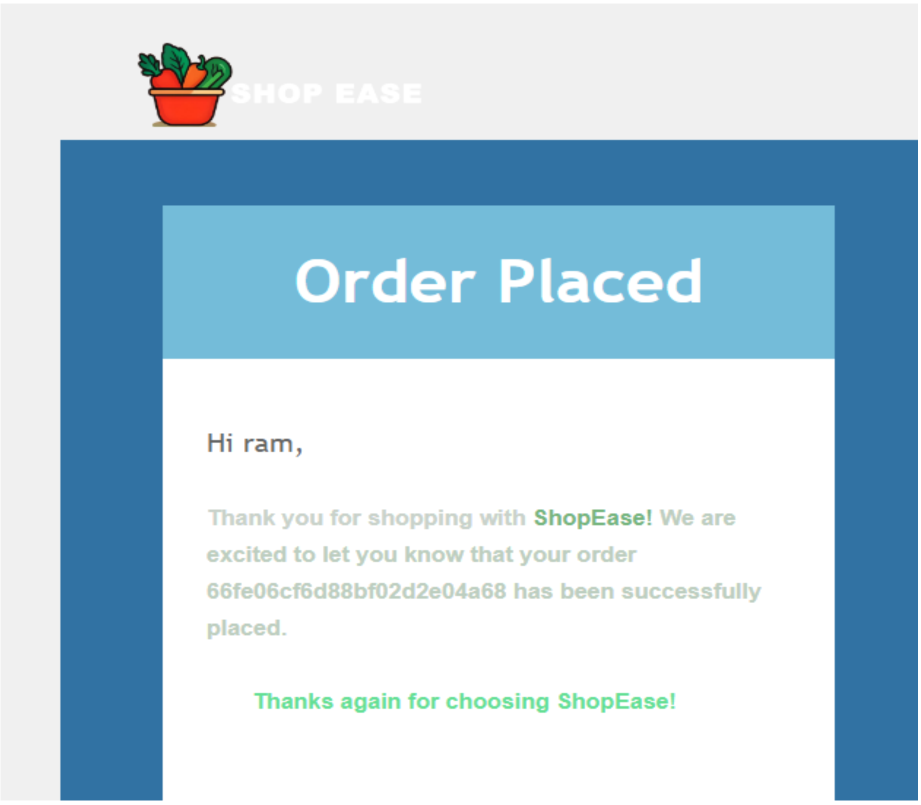


Figure V:E-mail template

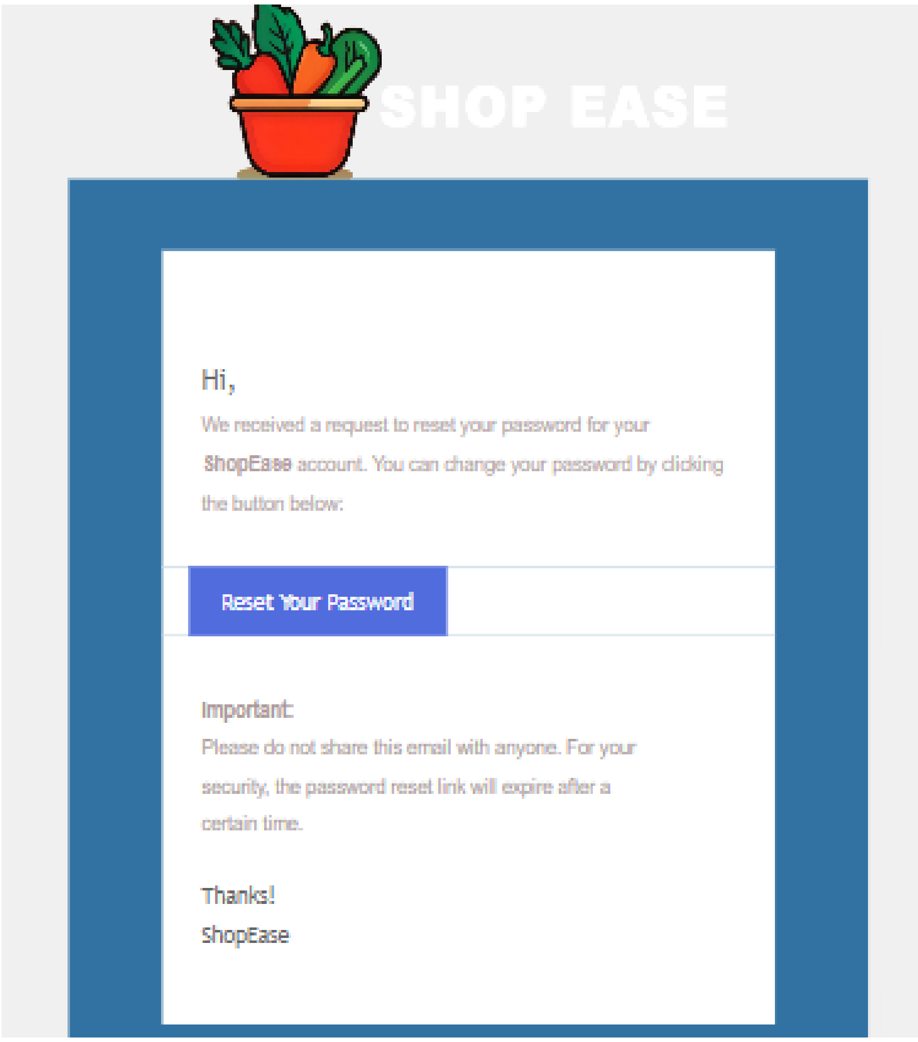


Figure VI:Reset password email-template